MODULE 06 · CHEAT SHEET

NVIDIA Ecosystem

Separuh depan: stack NVIDIA membuat LLM cepat & murah di-*serve* (GPU/Tensor Core, precision, quantization, TensorRT-LLM → Triton/Dynamo/NIM). Separuh belakang: Trustworthy AI memagari layanan itu (fairness, NemoGuard safety/topic, PII/UU PDP, grounding) lalu menjahitnya jadi service */ask* ber-audit. Ringkasan lima notebook (nb01–05).

Alur Modul 06: ① GPU + Precision + Quant → ② Serve: Triton→Dynamo→NIM → ③ Fairness + Tata Kelola → ④ Guardrails NemoGuard → ⑤ PII/UU PDP + Grounding → ⑥ Capstone /ask ber-audit

① GPU, Precision & Optimasi

- **Tensor Core**: hardware matmul; nyala hanya di precision rendah (FP16/FP8)
- **Precision** FP32→FP16: memori 6.18→3.10 GB (**2.0×**); matmul 34.81→6.23 ms (**5.6×**)
- **Quantization 4-bit** (NF4): 3.09→1.16 GB (**2.7×**) → 7B muat di T4 16 GB
- **TensorRT** = compiler umum; **TensorRT-LLM** = khusus LLM (paged KV-cache, in-flight batching, FP8/INT4)

‣ *perkecil + percepat sebelum deploy*

② Serving (Triton/Dynamo/NIM)

- **Triton** = baseline ujian: *dynamic batching*, multi-framework, metrics (`config.pbtxt`)
- **Dynamo** = flagship terkini (GTC 2025): disaggregated serving, KV-cache routing, multi-node
- **NIM** = model siap-pakai, kontrak **OpenAI /v1**; hosted vs self-host
- "one client, many backends": ganti `base_url+model` saja; *streaming* & request paralel naikkan throughput

‣ *menjalankan ≠ menyajikan*

③ Fairness & Tata Kelola

- **Demographic parity**: tingkat persetujuan sama antar kelompok?
- **Equalized odds** (lebih ketat): gap TPR/FPR antar kelompok; **disparitas < 0.1** = cukup adil
- Akurasi total menyembunyikan bias → fairness metrics membongkar per kelompok
- Teks bebas: **LLM-judge** NIM (sinyal saja). Deliverable: **Model Card** + ethics checklist

‣ *audit offline, bukan rail runtime*

④ Safety & Guardrails (NemoGuard)

- **Content Safety** `nemoguard-8b-content-safety`: taksonomi Aegis 23 kategori → JSON safe/unsafe
- **Topic Control** `nemoguard-8b-topic-control`: on-topic / off-topic (scope-limiter)
- **Jailbreak**: produksi = `/v1/classify` (Random Forest, bukan chat) → di sini **self-check** classifier Nemotron
- Input rails (jailbreak→safety→topic) *fail fast*; output rail cek jawaban juga

‣ *satpam masuk & resepsionis keluar*

⑤ Privasi (PII/UU PDP) + Grounding

- **PII masking** regex deterministik → `[NIK]`/`[PHONE]`/`[ACCOUNT]` (NIK 16-digit dicek dulu)
- **Data minimization**: model hanya melihat placeholder, masuk & keluar
- **UU PDP** No. 27/2022: breach report **≤ 72 jam**, denda **≤ 2%** pendapatan tahunan
- **Grounding**: jawab hanya dari konteks + sitasi [n]; verifikasi grounded/ungrounded

‣ *privasi = hukum, bukan etika belaka*

⑥ Capstone: /ask Berpagar

- **FastAPI** `POST /ask`: input rails → RAG (NIM) generate [n] → output rails
- Tiap rail aktif dicatat di field `rails` → jejak **audit** (Transparency/Accountability)
- **TestClient** smoke: valid=200 grounded+cited; jailbreak/off-topic blocked; cacat=**422** Pydantic
- Backend bisa ditukar (NIM→Ollama→TensorRT) tanpa ubah rail; runbook = deliverable

‣ *Trustworthy AI = lapisan runtime*

⑦ Edge Deploy (Jetson)

- **TensorRT Edge-LLM**: safetensors → **ONNX** (portabel) → **engine TRT** (terikat GPU) → inference C++ di Orin
- **Engine tak portabel**: embed SASS+taktik per arch (sm_87)+versi TRT → engine T4 ditolak di Orin ⇒ build di perangkat target
- Gotcha: venv `torch-CPU 2.12` (bukan container); `EMBEDDED_TARGET=jetson-orin`; engine OOM → `maxInputLen 256 / KV 512` + drop_caches
- **Worth it?** 42.7 vs Ollama 40.4 tok/dtk (≈ sama) → model kecil: **Ollama menang**; TRT sepadan di skala produksi

‣ *ukur dulu — NVIDIA-native ≠ otomatis lebih baik*

Quick-Ref Ekosistem NVIDIA & Trustworthy AI (5 Notebook)

nb	Lapisan / Alat	Tujuan	Sertifikasi
01	Tensor Core · FP16/FP8 · quantization 4-bit · TensorRT-LLM	Perkecil & percepat model untuk inference	Software Dev
02	Triton → Dynamo → NIM (Nemotron, OpenAI /v1)	Sajikan model ke banyak pengguna; streaming, throughput	Software Dev
03	Fairness metrics · LLM-judge · Model Card	Audit adil & tata kelola (offline)	Trustworthy AI
04	NemoGuard content-safety + topic-control · self-check jailbreak · PII · grounding	Pagar runtime tiap pesan; privasi UU PDP	Trustworthy AI
05	FastAPI /ask + semua rail · TestClient · runbook	Service ber-audit yang siap deploy	Software Dev + Trustworthy AI

Istilah Kunci

Tensor Core — Unit hardware NVIDIA untuk matmul precision rendah (FP16/FP8); kunci kecepatan inference LLM.
Quantization — Simpan bobot di bit lebih sedikit (4-bit NF4) → model besar muat di GPU kecil; di sini untuk inference, bukan training.
TensorRT-LLM — Compiler LLM NVIDIA: paged KV-cache + in-flight batching + FP8/INT4; mesin di balik NIM.
Triton / Dynamo — Server inference NVIDIA: Triton = baseline ujian (dynamic batching); Dynamo = penerus flagship GTC 2025 (LLM datacenter).
NIM — NVIDIA Inference Microservice; model siap-pakai (mis. `nvidia/nemotron-3-nano-30b-a3b`) lewat kontrak OpenAI /v1. Nemotron = reasoning model; matikan thinking via `extra_body chat_template_kwargs.enable_thinking=False`.
Fairness metrics — Demographic parity & equalized odds (gap TPR/FPR); ambang disparitas < 0.1; deterministik, beda dari LLM-judge.
Model Card — "Label gizi" model (data, tujuan, batasan) → memenuhi pilar Transparency.
NemoGuard — Model penjaga NVIDIA via NIM: content-safety (Aegis 23-kat) & topic-control (on/off-topic).
Content Safety / Topic Control — Rail keamanan (apakah berbahaya?) vs rail lingkup (apakah di dalam topik?).
PII masking — Ganti data pribadi (`[NIK]`/`[PHONE]`/`[ACCOUNT]`) sebelum ke model/log; data minimization.
UU PDP — UU No. 27/2022 Pelindungan Data Pribadi: breach ≤ 72 jam, denda ≤ 2% pendapatan tahunan.
Grounding — Rail transparansi: jawaban bersandar pada konteks + sitasi [n], diverifikasi grounded/ungrounded.
Capstone /ask — FastAPI service yang menjahit semua rail + RAG; keputusan diaudit lewat field `rails`.